HDOC - 70th day

DIGITAL ROOT USING RECURSION

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Problem Statement

Write a recursive function to calculate the digital root of a number. Use a static variable to count the number of recursive calls.

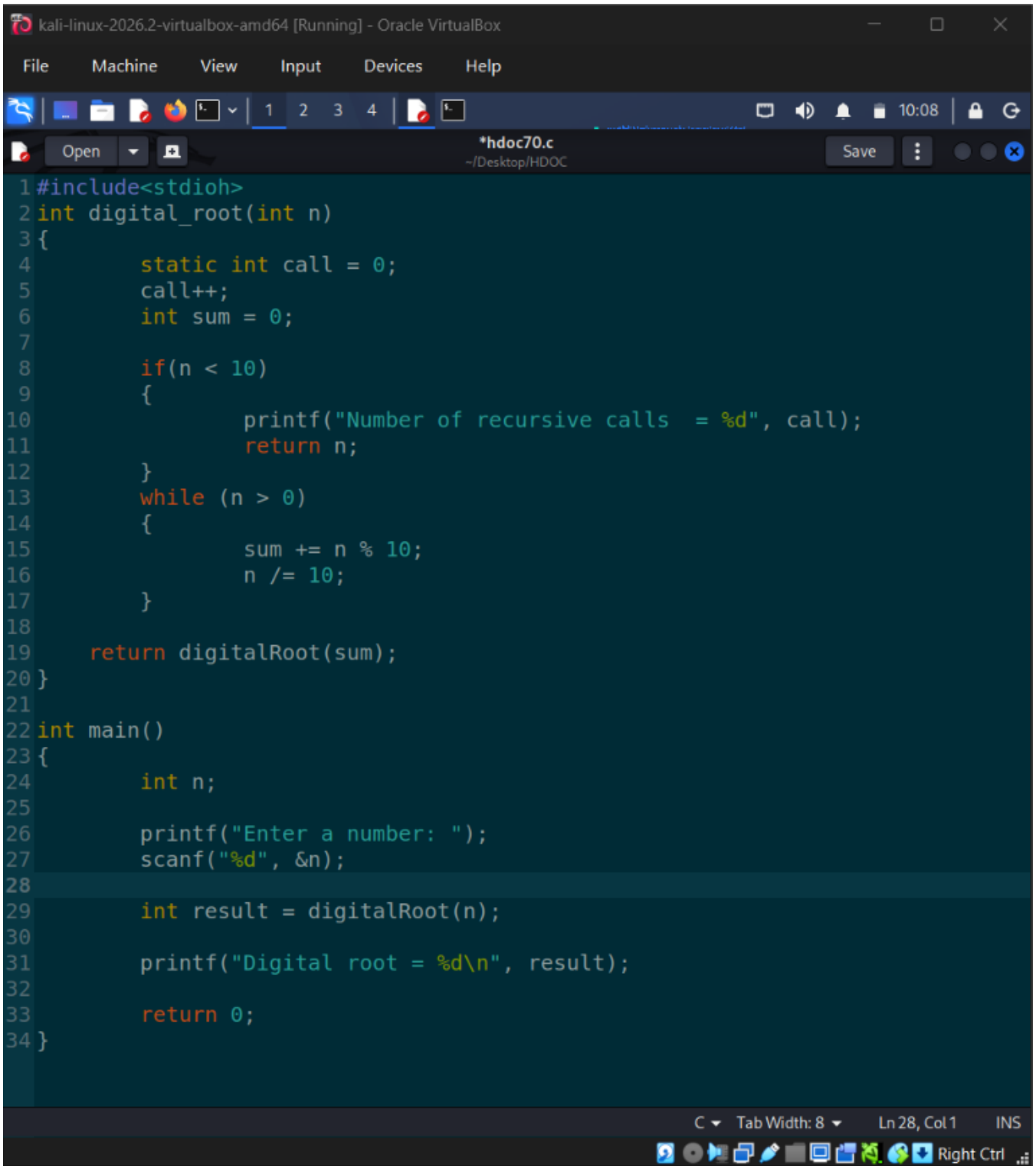
Algorithm

1. Start.
2. Read an integer n.
3. Call the recursive function digitalRoot(n).
4. Declare a static variable call and increment it on every function call.
5. If n < 10, display the number of recursive calls and return n.
6. Otherwise, calculate the sum of all digits of n.
7. Recursively call digitalRoot(sum).
8. Display the digital root.
9. Stop.

Test Case Table

S. No.	Input Number	Expected Output	Actual Output	Result
1	2589	Calls = 3 Digital root = 6	Calls = 3 Digital root = 6	Pass
2	211	Calls = 2 Digital root = 4	Calls = 2 Digital root = 4	Pass
3	6767	Calls = 3 Digital root = 8	Calls = 3 Digital root = 8	Pass
4	1001	Calls = 2 Digital root = 2	Calls = 2 Digital root = 2	Pass

C Program


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
10:08
Open
*hdoc70.c
~/Desktop/HDOC
Save
#include<stdioh>
int digital_root(int n)
{
static int call = 0;
call++;
int sum = 0;
if(n < 10)
{
printf("Number of recursive calls = %d", call);
return n;
}
while (n > 0)
{
sum += n % 10;
n /= 10;
}
return digitalRoot(sum);
}
int main()
{
int n;
printf("Enter a number: ");
scanf("%d", &n);
int result = digitalRoot(n);
printf("Digital root = %d\n", result);
return 0;
}
C Tab Width: 8 Ln 28, Col 1 INS
Right Ctrl

Compilation and Execution

Compilation command: gcc hdoc70.c -o hdoc70

Execution command: ./hdoc70

Execution Screenshot:

